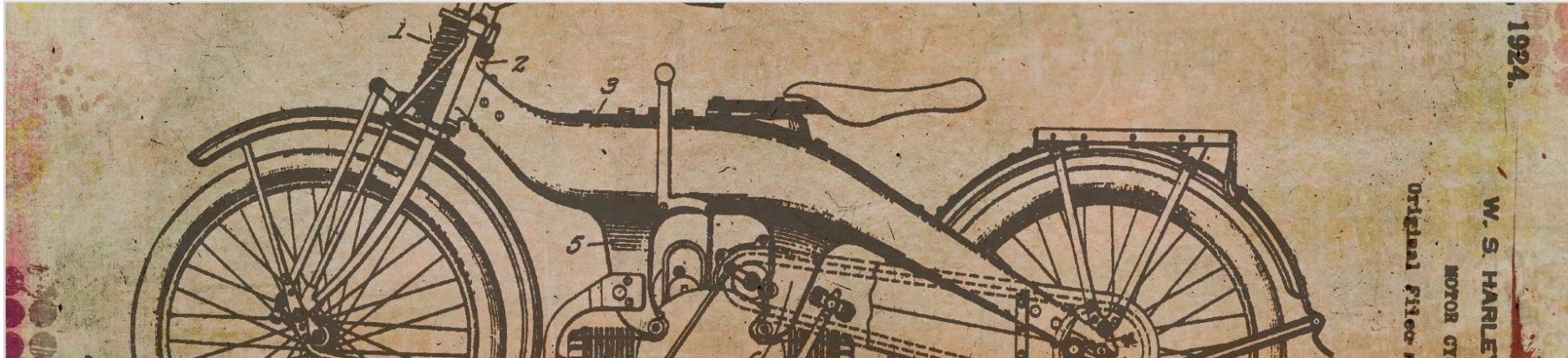


Intellectual Property & Technology Push

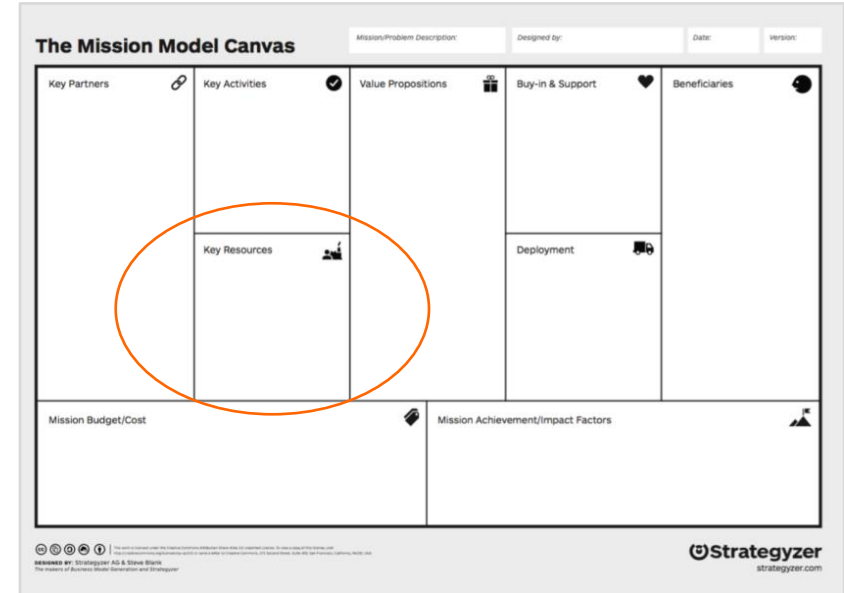
Prof. Dr. Orestis Terzidis



PATENTS

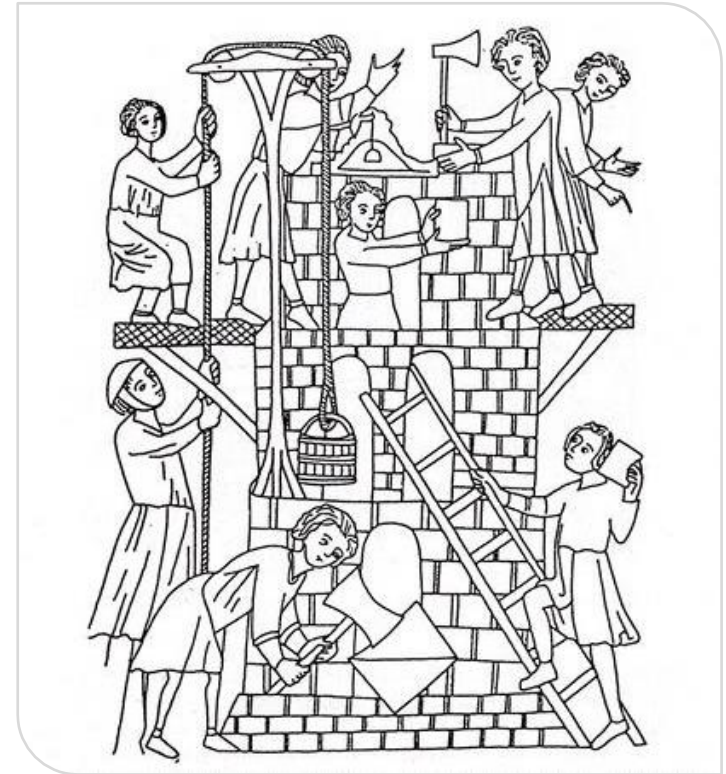
Resource-based Opportunities

- Do we have access to rare and valuable resources?
- How can we build a value proposition on the basis of these resources?
- Who is our customer for such a value proposition?
- One specific resource is new knowledge and related patents (or other forms of intellectual property).



Objective of Intellectual Property

- The objective of intellectual property law is to "**Promote progress**". (U.S. Const., art. 1, sec. 8, cl. 8.)
- By exchanging **limited exclusive rights** for **disclosure of inventions and creative works**, society and the patentee/copyright owner mutually benefit, and an incentive is created to create and disclose their work.
- Contrast: Medieval Guilds.



What is a patent?

The patent holder is the only authorized person to use the patented invention (§ 9 PatG).

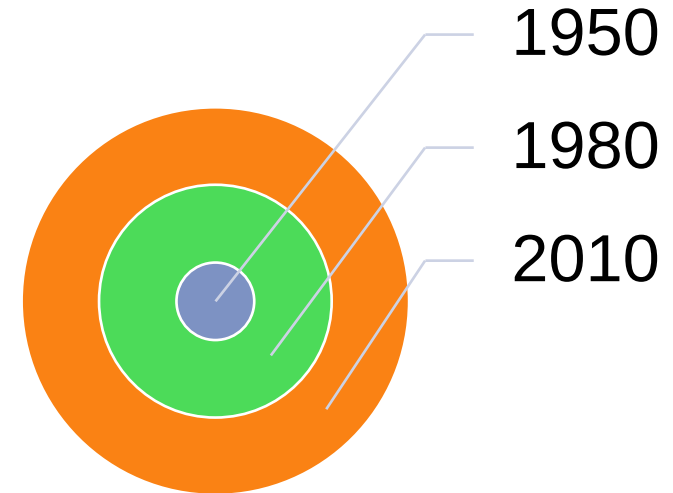
A patent is an exclusion right of limited duration, which the state grants for the disclosure of an invention.

Novelty	An invention is considered new if it is not state of the art (§ 3 sentence 1 PatG).
Inventive activity	An invention requires an inventive step; it may not be obvious for an expert skilled in the art (§ 4 PatG).
Industrial application	An invention needs to have an industrial application (§ 5 sentence 1 PatG).



State of the Art

- State of the art refers globally to the highest level of development, as of a technology, device, technique, or scientific field achieved at a particular time.
- It is a growing body of knowledge.
- The patent authorities explore the SoA mainly through:
 - International patent databases (see e.g. www.dpma.de)
 - International scientific publications
 - Commercially available products



Patent Application

Filing

Examination

Infringement

Heading & Summary

- Maximum of 1,500 characters
- Name of the inventor & the invention, abstract of the technology

Full Description

- Extensive description of the invention, function, potential uses

Key Section: Patent Claims

- All attributes essential for realization of the invention
- Abstract formulation to ensure maximal protection

Drawings

- Designs with explanations of the components in the description

Inspection Process



Formal Examination

- Are the documents filled out correctly?
- Is the technology part of patent exclusions?

Research (Optional)

- To which degree can the invention be protected?

Examination

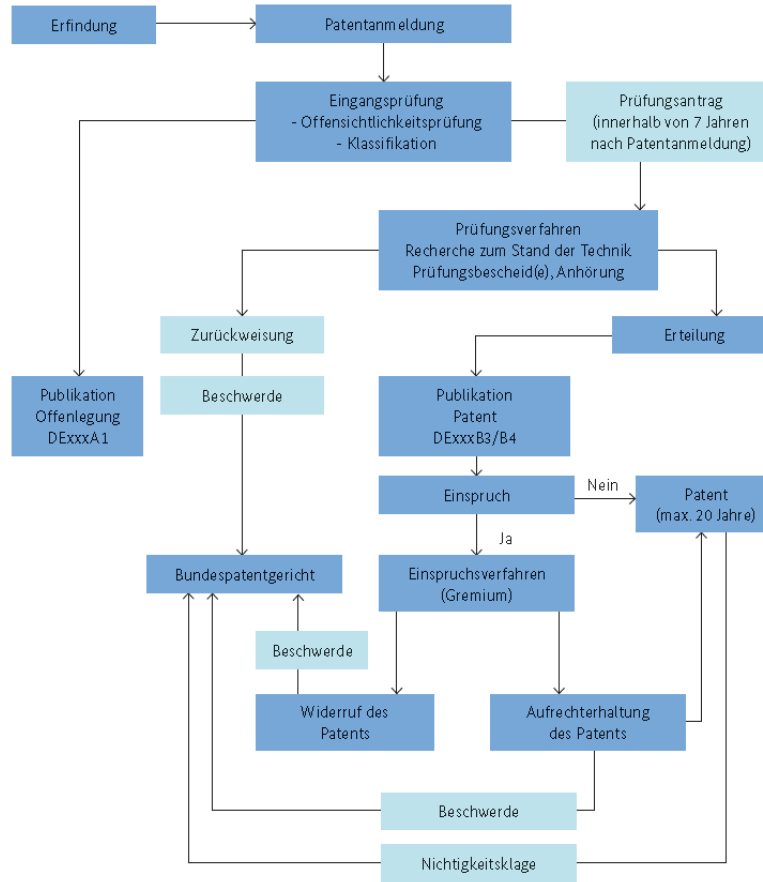
- Does the invention meet the criteria novelty, inventive step and industrial application?

Disclosure of Patent Application

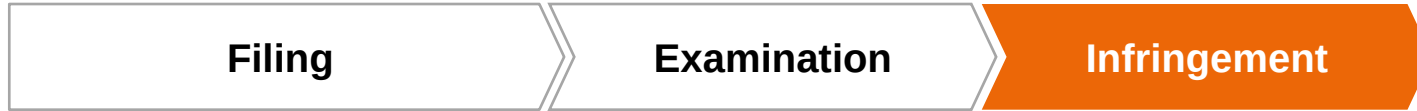


- The disclosure happens automatically after 18 months.
- Thereafter, other companies can find the patent application and its technology.
- This happens whether the patent is finally granted or not.
- Competitors can file an opposition against the patent in the first nine months after the granting of the patent.

Patent Examination Process in Germany



What happens in case of a violation?



Patent infringement lawsuit

- In regular court, not patent office.
- For the trial duration the patent is valid.

Claims

- Injunctive relief (Unterlassungsanspruch)
- Claim for indemnity (Entschädigungsanspruch) on basis of
 - the lost profit,
 - the profit the infringing party made or
 - costs for a license

Examples

Filing

Examination

Infringement

Patent infringement: Apple has to pay
532,9 million us dollars

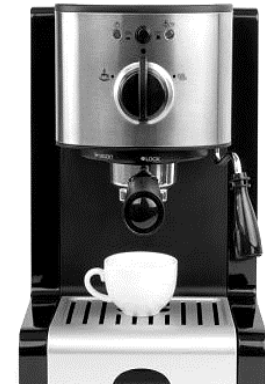
Apple has to pay smart flash 532.9 million us dollars for a triple patent infringement by order of a Texas jury. The plaintiff's claim, however, was significantly higher and Apple wants to appeal this ruling.

Coffee capsule dispute between
Nespresso and Ethical Coffee Company
to court

Patent infringement in France
Coffee capsule dispute: Nespresso competitor claims
150 million euros

Zurich. The coffee capsule dispute between market leader Nespresso and its competitor ethical coffee company (ECC) that has been raging for years ends up in court again. ECC believes one of their patents to be infringed by Nespresso as they are using a special technique for themselves now.

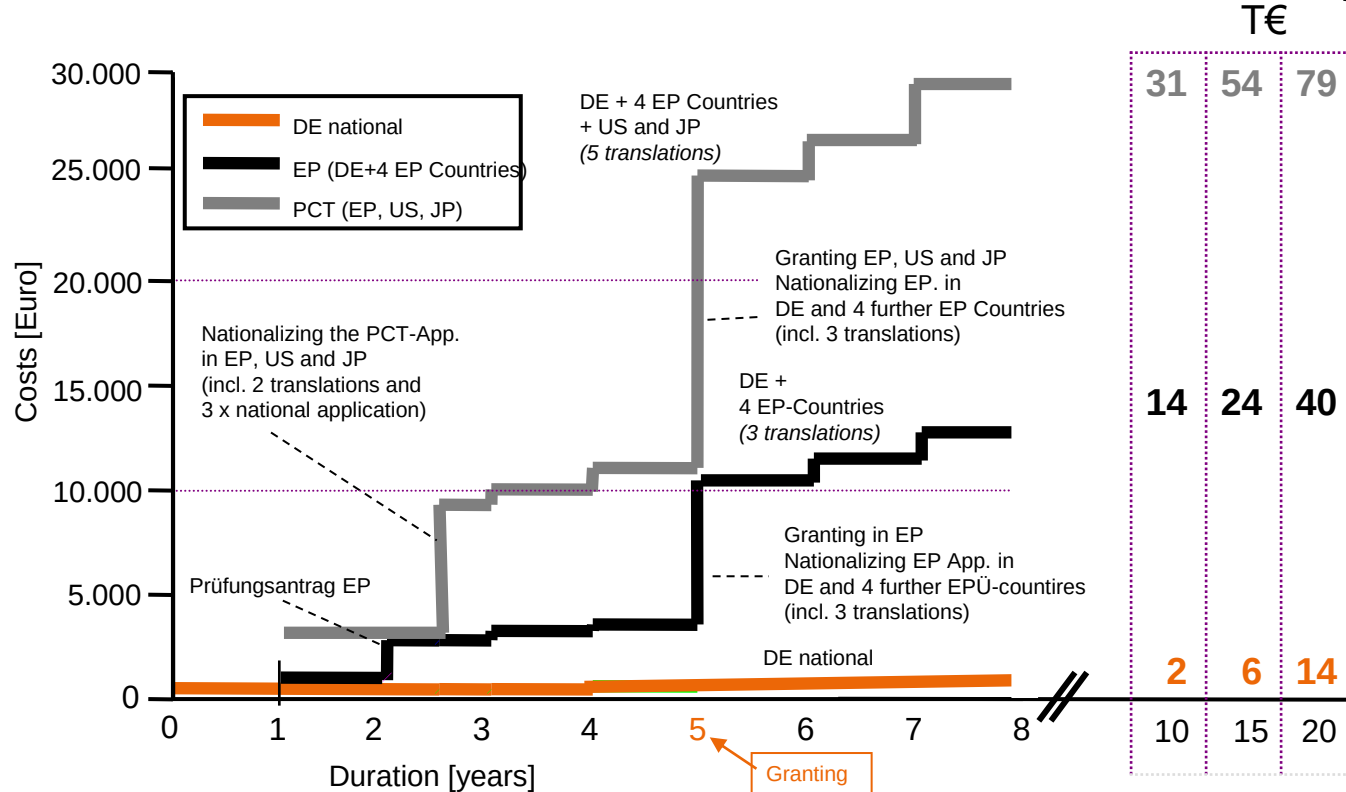
Nespresso und dem Konkurrenten Ethical Coffee Company (ECC) landet erneut vor Gericht. ECC sieht ein eigenes Patent verletzt, Nespresso nutze eine spezielle Technik nun für sich.



Costs from patent application to granted patent

Type of invention	Examples	Costs of patent office and patent attorney
Extremely simple	Electric switch; coat hanger; paper clip; diapers; earmuffs; ice cube tray	6.500 € - 8.300 €
Relatively simple	Board game; umbrella; retractable dog leash; belt clip for cell phone; toothbrush; flashlight	8.300 € - 10.100 €
Minimally complex	Power hand tool; lawn mower; camera; cell phone; microwave oven	10.100 € - 11.000 €
Moderately complex	Ride-on lawn mower; simple RFID devices; basic solar concentrator	11.000 € - 13.250 €
Relatively complex	Shock absorbing prosthetic device; basic to moderate software or systems	13.250 € - 15.500 €
Highly complex	MRI scanner; PCR; telecommunication networking systems; complex software or systems; satellite technologies	> 15.000 €

Cumulated External Costs for a Patent Family



Exclusions

In Germany a few things are excluded from patenting:

- Business ideas, plans and regulations
- Plant and animal species
- Unethical inventions
- Some inventions in microbiology, biotechnology and software can be patented

Furthermore patent protection is typically not possible for:

- Substitution of a material (e.g., plastic for metal)
- Change of size/form of an existing machine/component
- Making something more transportable
- Substitution of one element with a similar one

IP Strategy

Opportunities

- Control over patent and invention
- Competitors cannot apply for the same patent: monopoly
- Reputation, Innovation, and higher valuation
- Communication & cooperation with partners
- Monitoring competitors through their patents

Risks

- Disclosure of the invention after 18 months
- Application process expensive
- Maintenance expensive due to annual fees
- Enforcing patent infringements involves high legal expenses and lawyer costs

First Patent, then Publish

Examples for Publication:

- *Journal articles/paper, preprint, abstract, Proceedings*
- *Presentation, lecture, conference poster*
- *Dissertation, master thesis, public seminal paper*
- *Internet: Homepages, Emails, Chats, ...*

Date of a publication = day when the content is made publicly accessible
→ **Accessibility** is decisive, *not proven access*

Public = an unmanageable circle of people
No publication, if a **Non Disclosure Agreement (NDA)** was signed

Internationalization Strategies

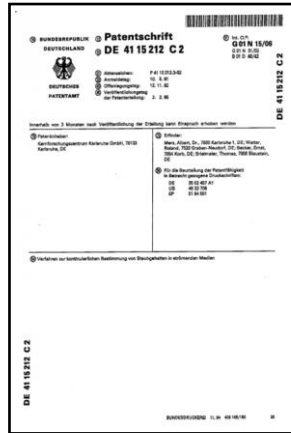
Criteria for the selection of countries

- Where are the main **customer markets**?
 - Strategically select markets in a way that the **remaining markets** are not attractive for competitors.
→ E.g.: Germany, France, UK, Italy USA, China, Japan
- Where are potential licensees and their competitors?
- Possibilities of enforcement in case of patent infringement
 - Possibilities of monitoring
 - Effectiveness and efficiency of legal system

PCT & Patent Families

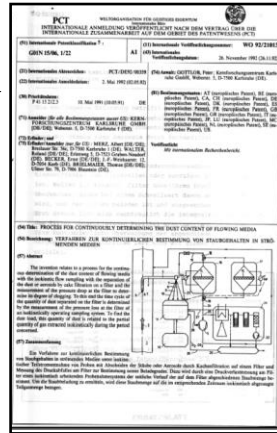
Priority
Application

DE



Subsequent
Application

PCT



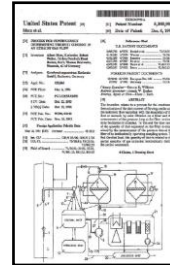
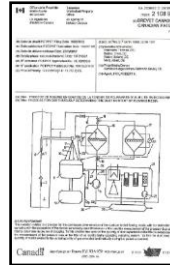
> 30
Countries in
Europe

EP

JP

CA

US



The **Patent**
Cooperation Treaty
(**PCT**) is an
international **patent**
law treaty,
concluded in 1970.

It provides a unified
procedure for filing
patent applications
to protect inventions
in each of its
contracting states.

Who owns the Invention of an Employee?

- Who owns the invention? The inventor or his legal successor → § 6 PatG (German Law)
- Rights and duties of an employed inventor towards his employer (e.g. KIT, company):
 - German „Arbeitnehmererfindungsgesetz“ (employee invention act)
 - Relevant for patentable inventions (patents or utility models)
 - Employer inventions (Dienstserfindungen) are inventions, that
 - *Occurred during the regular professional activity or*
 - *Are primarily based on experiences and knowledge acquainted in the professional context*
 - All other inventions of the employee are free inventions
 - Mandatory Disclosure (Meldepflicht) of the employee towards the employer

Figures at the KIT

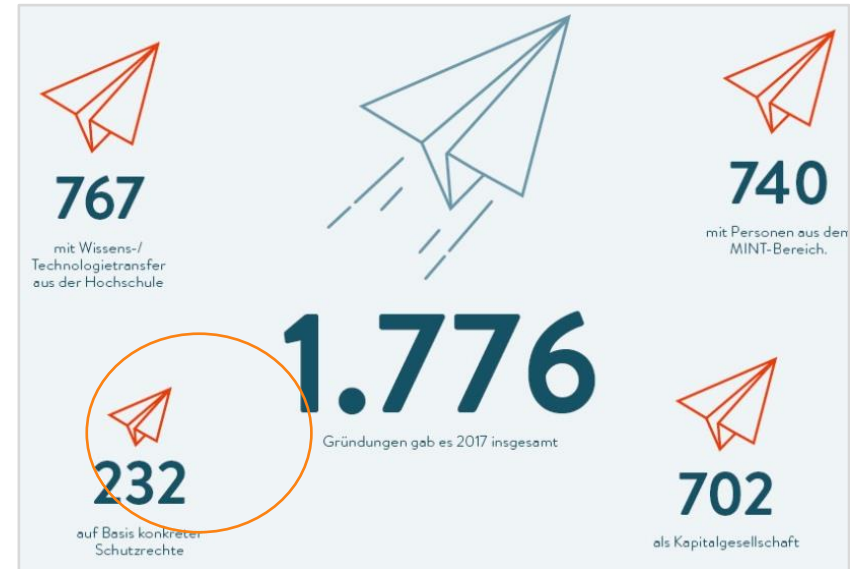
Per Year	Overall
Invention Disclosures	~150
Patents / Utility Models	~60

Actuals	Overall
Patents / Utility Models	~1800
Licensed Patents / Utility Models	~1300
Brands / Design Patents / Designs	~50

IP IN STARTUPS

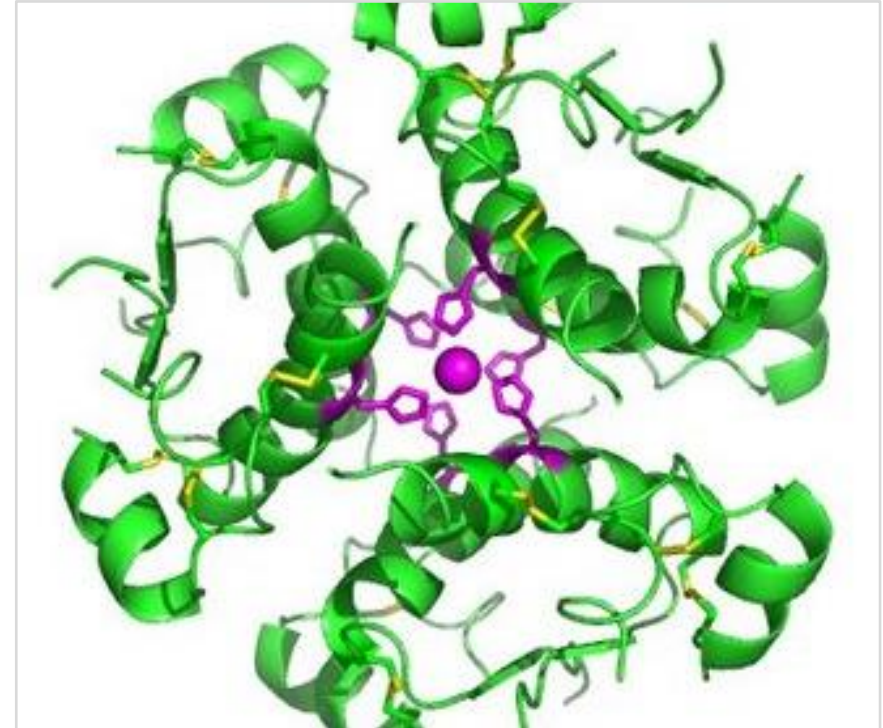
13% IP-based Ventures at German Universities

- A **New Technology-Based Firm** (NTBF) is an entrepreneurial organization with the goal to actively create, develop, and/or commercialize offerings based on technology and/or research, particularly innovative products, processes, applications and services, which is no more than 12 years in operation.
- A special case of an NTBF is the **Research-Based Startup** (RBSU). It is a new firm based on the research findings at a university or public research institution.



The Role of Patents for VC Financing

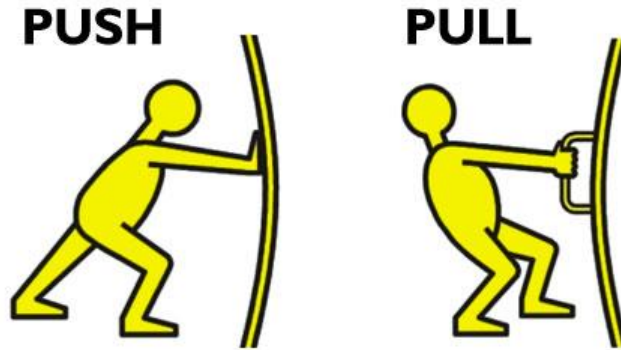
- Entrepreneurs have to convince investors of the growth potential of their company. The quality of a venture is often not directly observable, but patents may be an indicator.
- While preparing a patent application is costly and requires the disclosure of private information, **patents have an important signaling value** (Häussner et al., 2012).
- While patents affect the venture capitalists' decision making in their property rights function, **we find no indication that they serve as technology quality signals** (Hoenig & Henke, 2015).



Startup Success Factors

- Meta-Analysis of 31 studies
- 11,259 NTV between 1991 and 2000 in the USA
- After 5 years ~22% with > 5 FTE still existed
- Out of 24 mentioned success factors, 8 were significant.
- Supply chain integration
- Market scope
- Firm age
- Size of founding team
- Financial resources
- Founders' marketing experience
- Founders' industry experience
- Existence of patent protection

Song et al. (2008)



TECHNOLOGY PUSH & MARKET PULL

Technology push...

... is the **development** and market **introduction** of a **new technology-based product** or service initiated **by new technologies** rather than **customer needs**.

The result of a technology-push strategy may become a **radical innovation** creating a **new market** with **high profit** potential.

Market Pull...

... is the **development** and market **introduction** of a **new product** or service induced by **customer demand**.

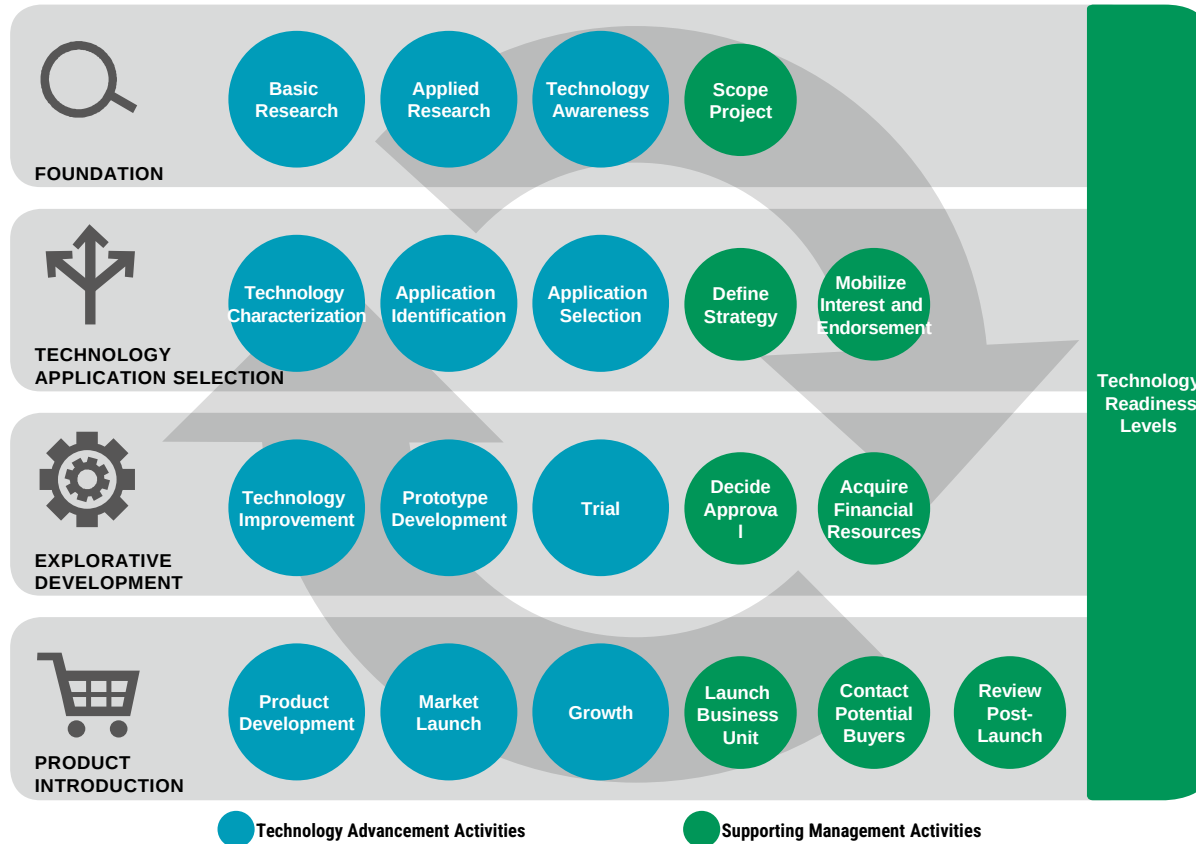
This requires the identification of **latent unsatisfied customer needs**.

The needs can be identified by **market research** with appropriate instruments, e.g. customer surveys or qualitative methods (including 'design thinking').

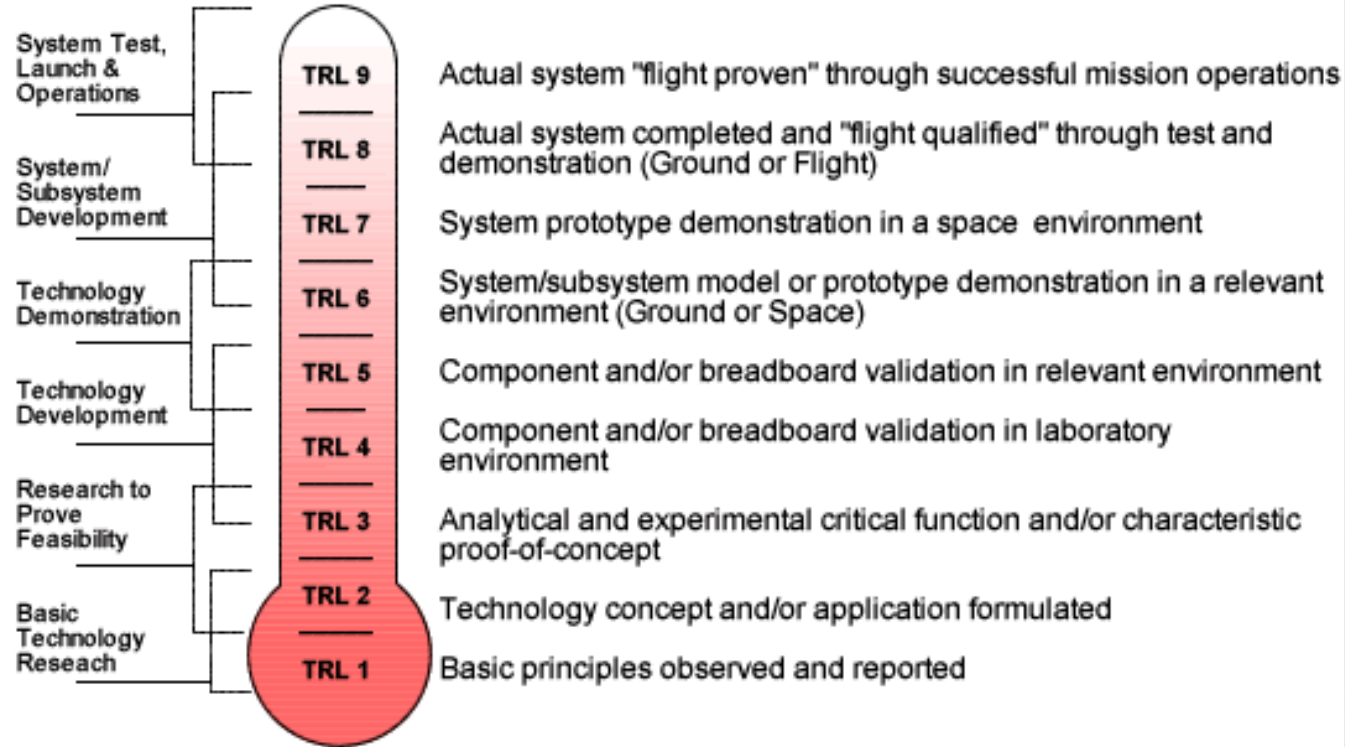
Technology push and market pull

Description/attribute	Technology Push	Market Pull
Technological Uncertainty	High	Low
R&D Expenses	High	Low
R&D Duration	Long	Short
Market-related uncertainty	High	Low
Time-to-market	Uncertain/unknown	Certain/known
R&D Customer Integration	Difficult	Easy
Kinds of Market Research	Qualitative	Quantitative
Need of change in customer behavior	Extensive	Minimal

TAS Framework



Technology Readiness Levels (TRLs)

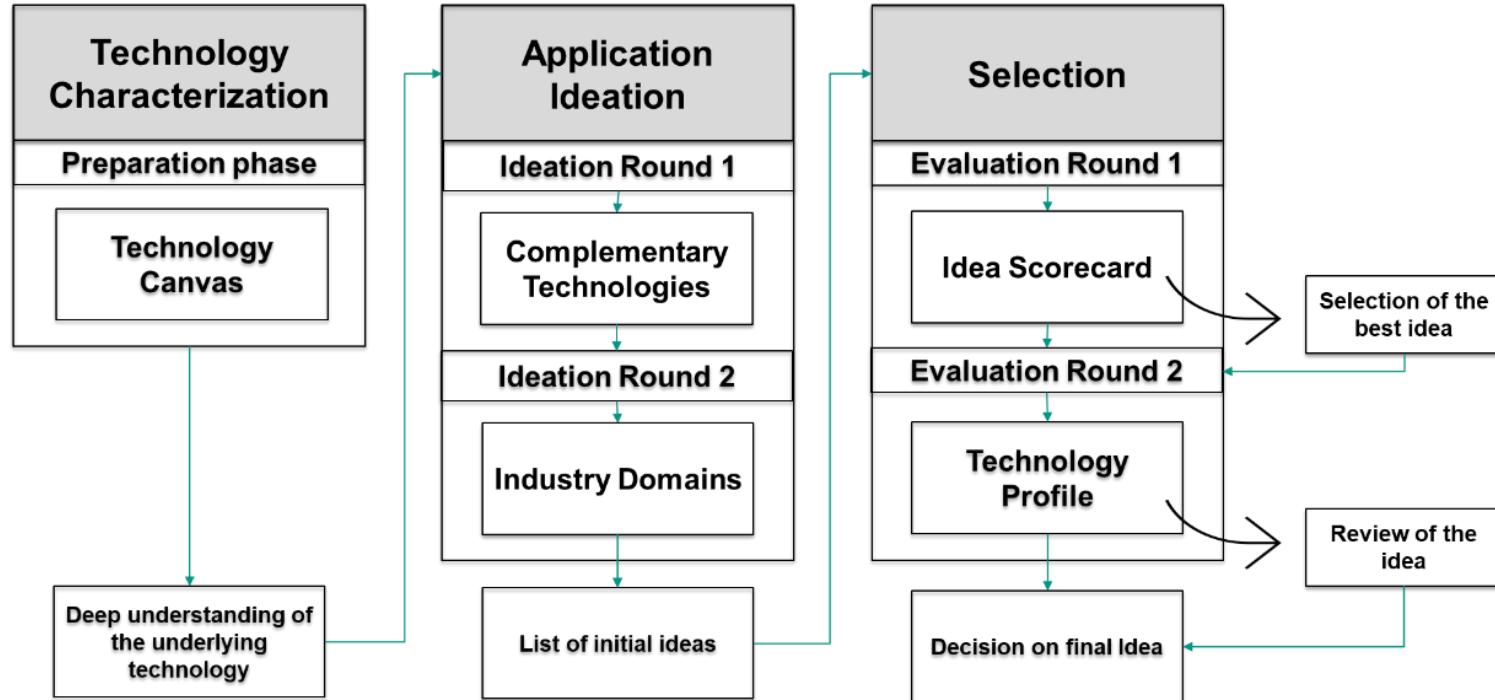


Technology Characterization

Application Ideation

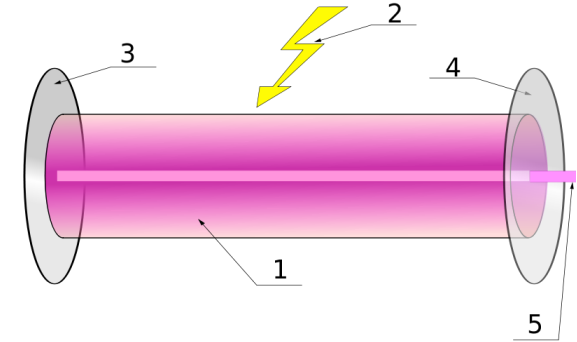
Selection





Technology

- A technology is an artifact that **performs a specific function** by transforming, transporting or storing energy, matter or information.
- It is implemented as a **specific systemic arrangement** (technical system).
- It refers to tools, machines, techniques, crafts, systems, and software.
- Technology is applied to solve problems & achieve goals.



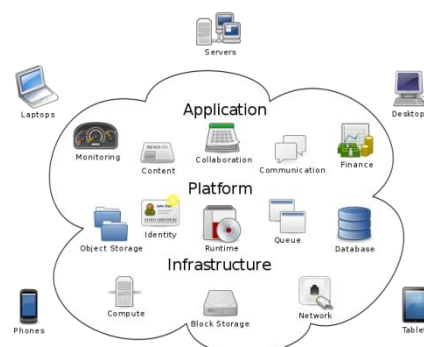
Components of a typical laser:

- 1 Gain medium
- 2 Laser pumping energy
- 3 High reflector
- 4 Output coupler
- 5 Laser beam

Technology Canvas

Title		Team Name
Example Application What is an example where the technology is used? 	Core Idea, Function Transformation, transportation, storage of energy, matter, information 	Promise & Novelty Potential benefits and what is new 
Current Practice What practices are used currently? What are limitations of current practices? 	Drawing, Structure Drawing related to the central concept, typically describing the structure 	Limitations What are limitations or weaknesses? 

Example: Cloud Computing

Title	Name	
Cloud-Computing	Example Application Web Shops Web Mail	Idea Description Dynamic deployment of scalable IT services through shared IT resources over networks. Real-time self-service based on Internet technologies. Many companies work on the same platform and share unassigned IT resources (multi-tenant / multi-tenancy)
State of the Art Each company installs a dedicated client-server environment (single-tenant architecture) The IT department of a company controls the provisioning and adjustment of IT performance "manually"	Drawing  The diagram illustrates the layers of cloud computing. At the top, 'Application' layer includes Monitoring, Content, Collaboration, Communication, and Finance. Below it is the 'Platform' layer with Object Storage, Identity, Runtime, Queue, and Database. The bottom layer is 'Infrastructure' containing Compute, Block Storage, and Network. Various client devices like Laptops, Desktops, Phones, and Tablets are shown connected to this central cloud. The entire cloud is labeled 'Cloud computing' at the bottom.	Promise & Novelty On-demand self-service. Scalable IT resources. Back-up services. Consumption-based billing
36		Limitations Concerns about Data Protection Trust in Provider is key

Example

Arrangement to produce fluid flows via Electroosmosis		# Penthouse	20.12.18
Example Applications • Oil Analysis • Online Analysis to Maintenance	Core Idea • Device for producing fluid flows & particle flows - small scale $< 5\text{mm}$ • Transport via electrical fields	Promis & Novelty • control fluid flows in any direction ↳ multidirectional flows • enables to - mix, divide - drain • multiple analysis	
Current Practise • many Lab-On-Chip appl. for medical analysis exist • pressure gradient induce flow	Drawing Structure 	Limitations • mass flow, size • viscosity, particle size • electricity available • not integrated in production process	

Application Ideation

- Ideation is the idea generation process through approaches such as Brainstorming, Brainwriting, sketching, and other techniques.
- Ideation is a key stage in an innovation project.
- The aim is to generate a **large quantity of ideas** that the team can then filter and select the best ones in order to design a value proposition.



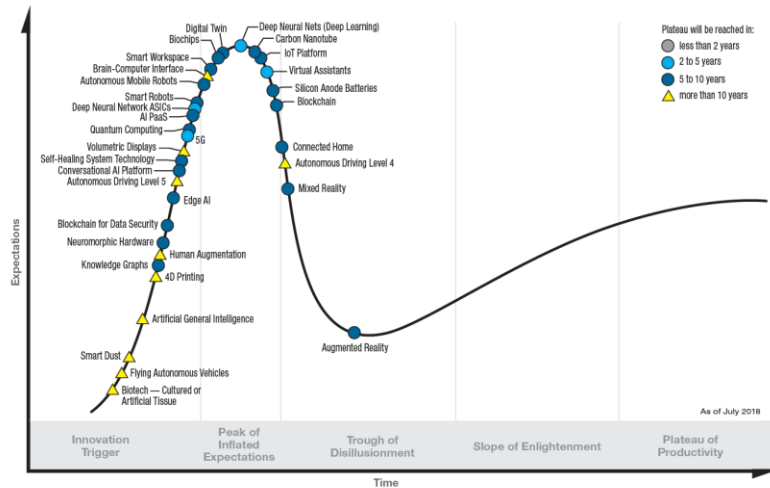
Facilitate fruitful ideation

- Base the ideation on the compiled information that you gained in the technology characterization.
- Create diverse perspectives and foci.
- Step beyond the obvious to discover also unexpected potentials of your technology.
- Use the team diversity to inspire each other.
- Create volume and variety in your application options.



Ideation anchors

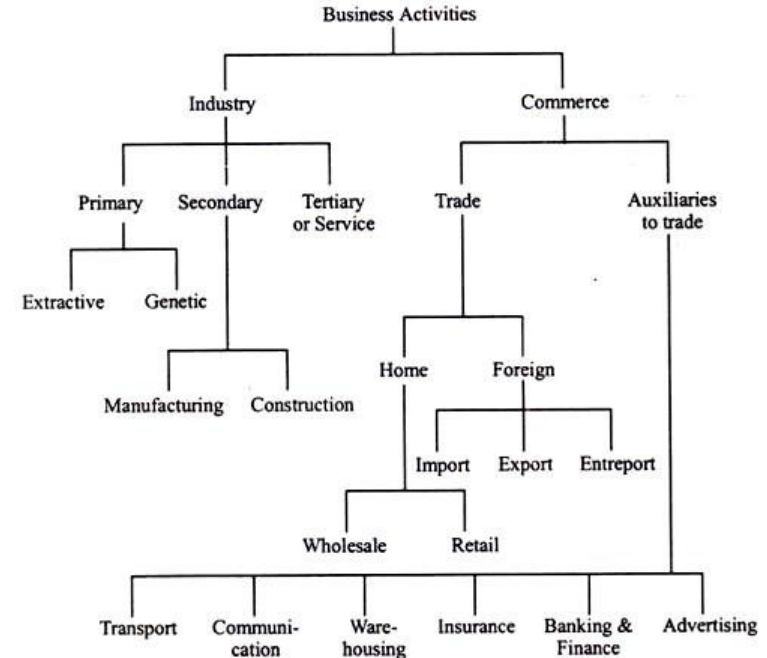
Hype Cycle for Emerging Technologies, 2018



gartner.com/SmarterWithGartner

Source: Gartner (August 2018)
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Gartner



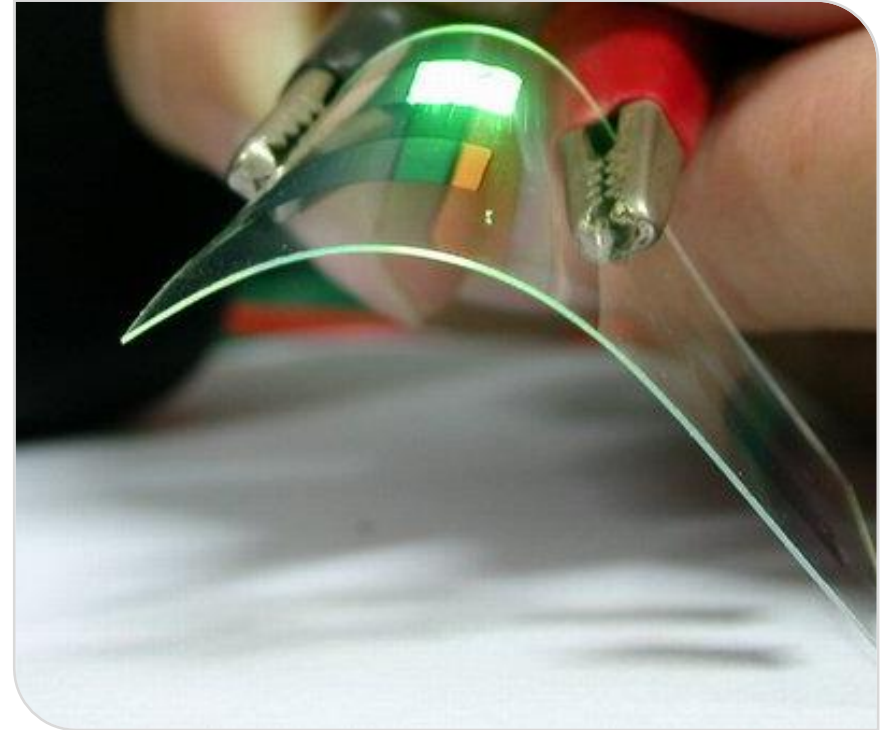
Selection

	Technical Feasibility	Market Potential	Profitability	Team Values Fit	Market Entry
Idea 1					
...					
...					

CASE STUDY: NOVALED

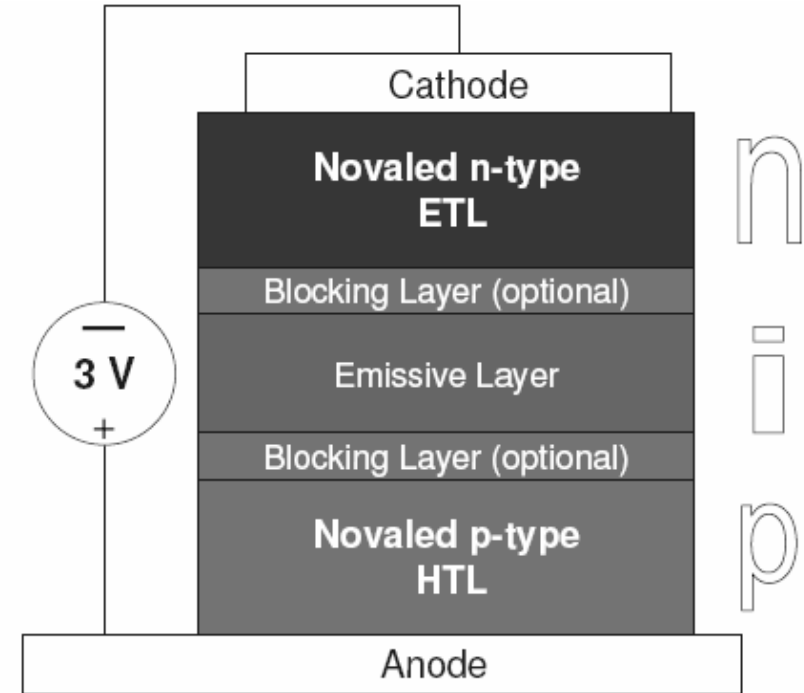
Novaled

- Novaled GmbH was founded in 2001 as spinoff of the TU Dresden with the scope to develop OLEDs.
- In 2013 CHEIL/Samsung acquired 50% of Novaled for 260m€.
- OLED was discovered in the 1950ies by André Bernanose at the University of Nancy in France.
- OLEDs are primarily used as lighting elements or in Displays.

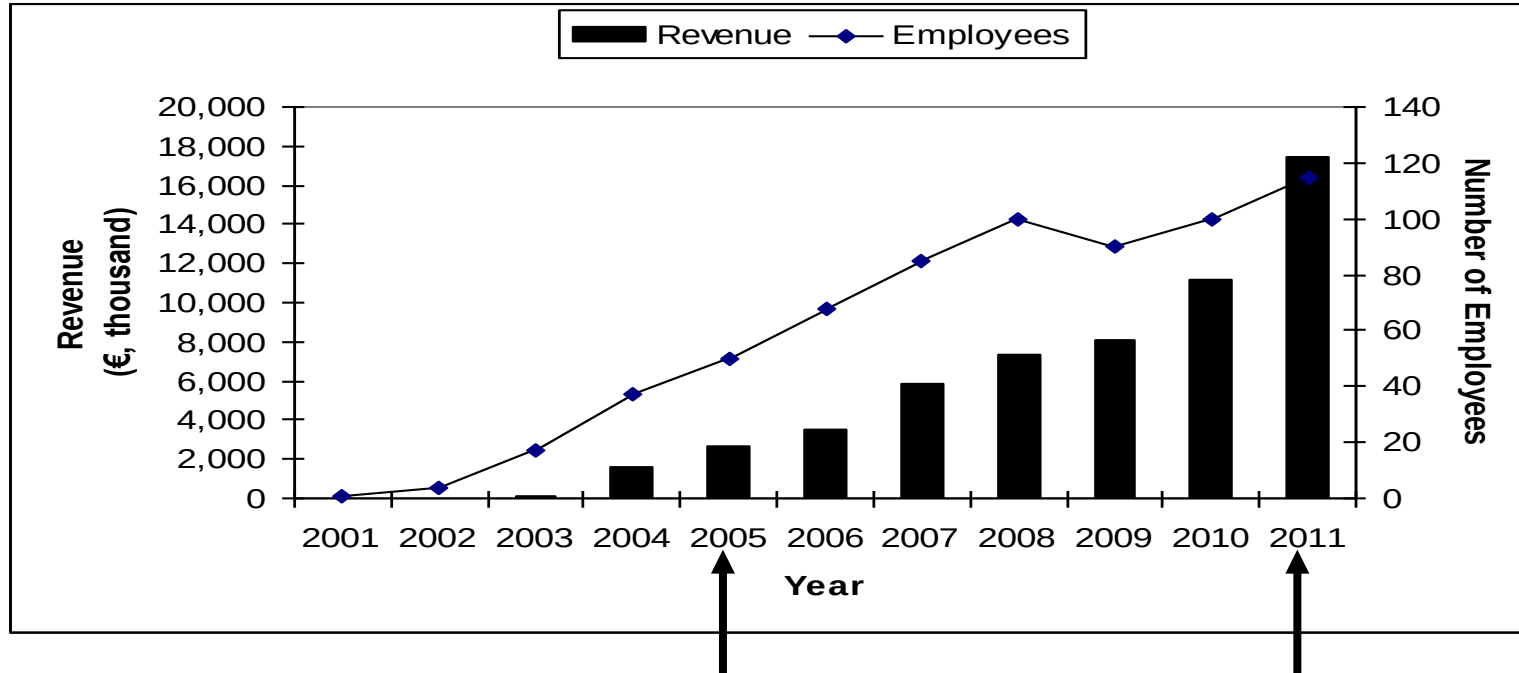


Novaled Technology: Solving Efficiency Problems

- Doping of the transport layer
- High efficiency OLEDs to require pin-structure (“PIN concept”)
- - p-type doped hole transport layer
- - intrinsic emission layer (may be emitter doped)
- - n-type doped electron transport layer.



Development of Novaled



130 patents

> 400 patents

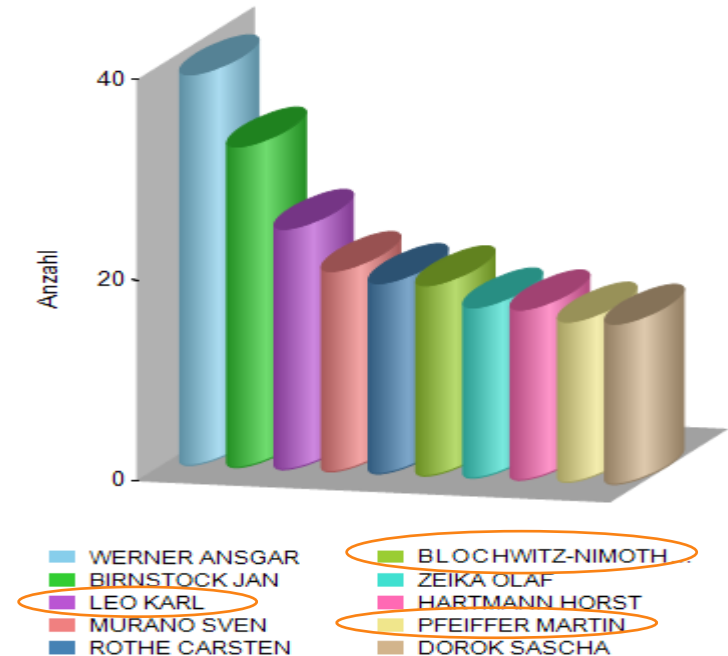
Partnership with TUD and Fraunhofer IAPP

- In 2003, Novaled agreed on a **framework contract** with TU Dresden **before the first investors came on board**.
- **TUD patents** were transferred to Novaled against **shares**.
- An agreement for a strategic **research partnership** TUD/Novaled and FHG-IAPP/Novaled was signed.
- Until 2010, **31 patent families** of the TUD were transferred.

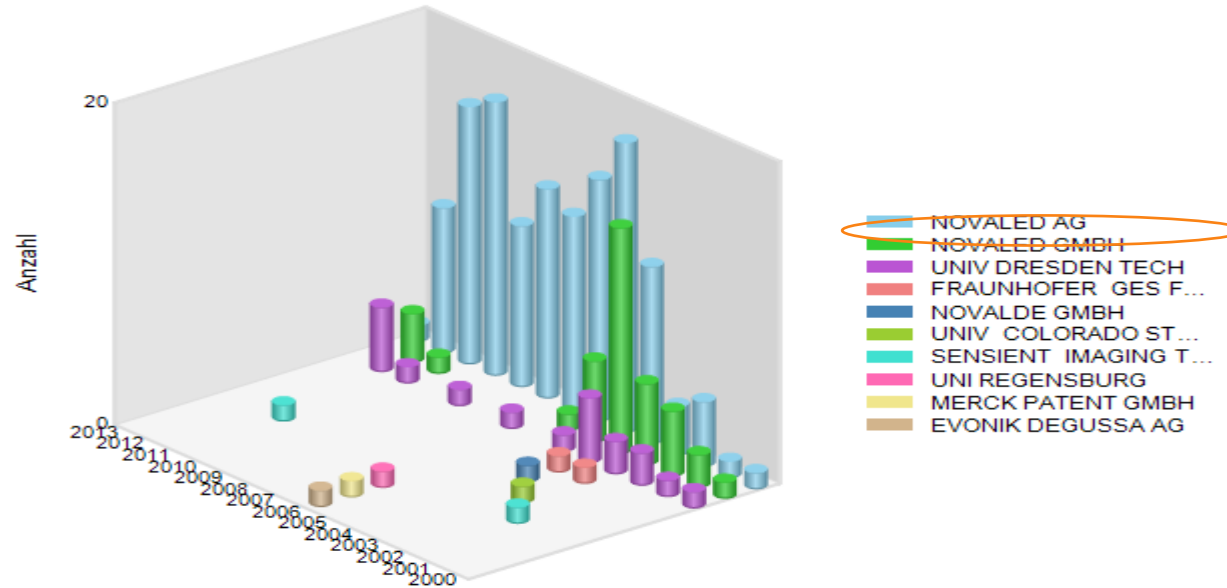


Novaed Patent Portfolio

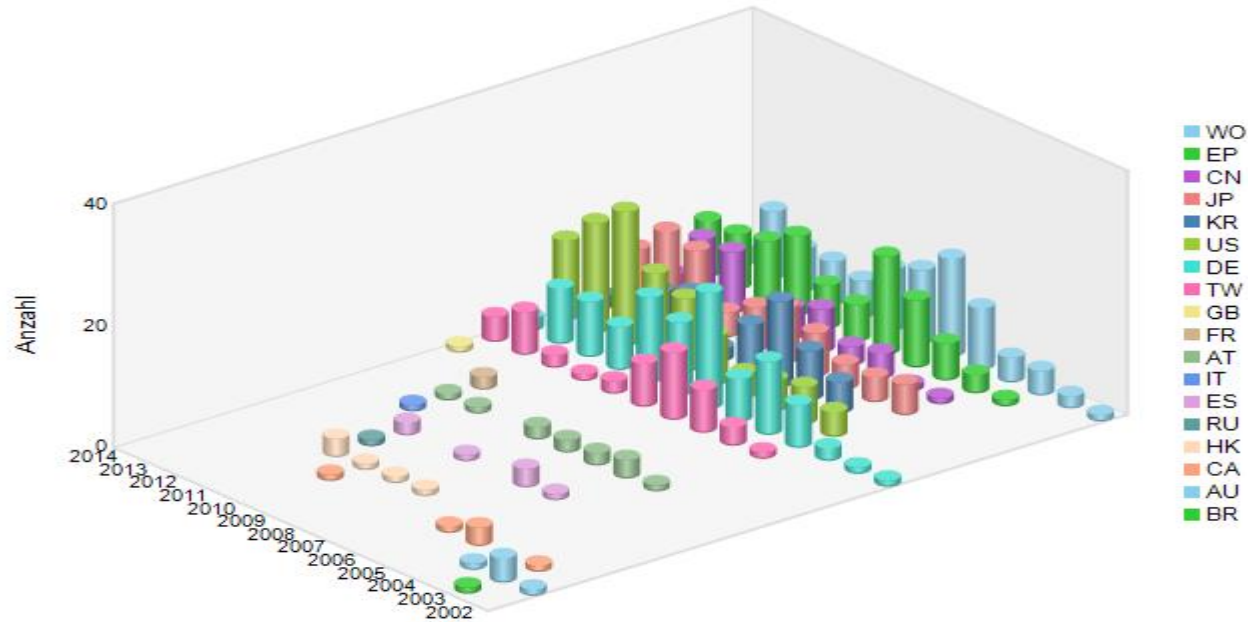
- The Novaed patent portfolio comprises **947 patents** and patent applications in **142 patent families**.
- 273 patents in 90 patent families were granted (status September 2014).
- Estimated overall investment of **5-15m€**. (annual costs of ~ 500 k€.)
- Adequate investment given the OLED **multi-billion Euro market**.



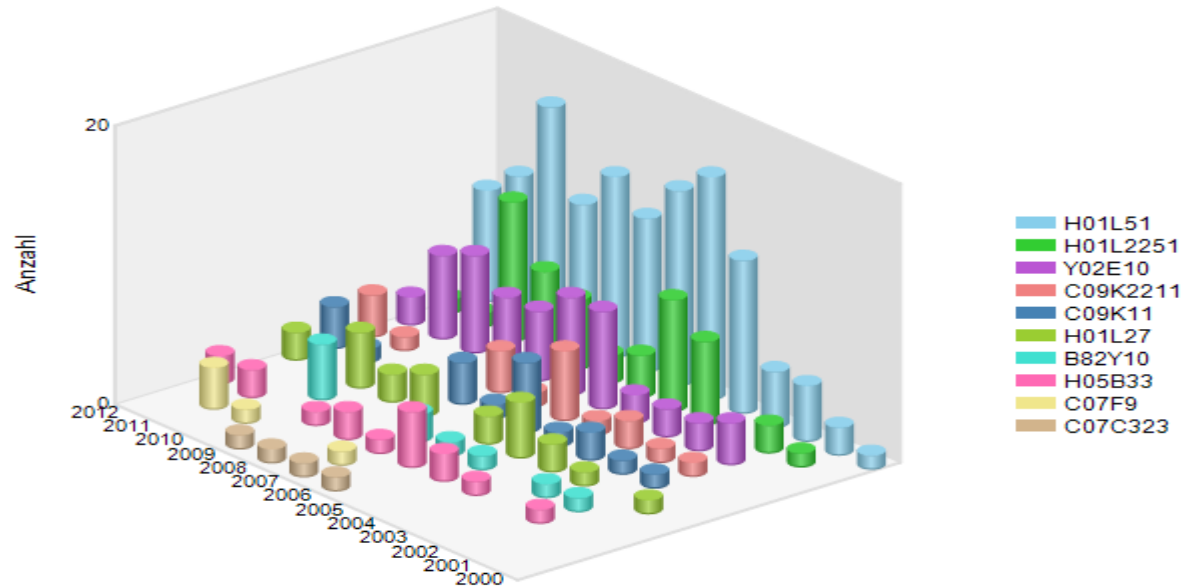
Partner Development over Time



Countries



Patent Classes



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Thank You!

